

Melakukan Analisis Menggunakan Python

Analisis Data Patient Kanker Mata

Kanker mata merupakan penyakit yang disebabkan oleh sel perusak yang terjadi dibagian mata. Data disediakan mengenai riwayat pasien, diagnosa pasien dan treatment pasien. Analisis data bertujuan untuk memberikan informasi mengenai kanker data. Data bersifat informatif dan Data Dapat dipahami dengan mudah.

Pemanggilan library

```
In [4]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Menginput data

```
In [5]: df=pd.read_csv("datasetEyeCancer.csv")
df.shape
```

```
Out[5]: (5000, 18)
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 18 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   baris                 5000 non-null  int64  
 1   Patient_ID           5000 non-null  object  
 2   Age                  5000 non-null  int64  
 3   Gender               5000 non-null  object  
 4   Cancer_Type          5000 non-null  object  
 5   Laterality           5000 non-null  object  
 6   Date_of_Diagnosis    5000 non-null  object  
 7   Stage_at_Diagnosis   5000 non-null  object  
 8   Treatment_Type       5000 non-null  object  
 9   Surgery_Status       5000 non-null  bool    
10  Radiation_Therapy    5000 non-null  int64  
11  Chemotherapy         5000 non-null  int64  
12  Outcome_Status       5000 non-null  object  
13  Survival_Time_Months 5000 non-null  int64  
14  Genetic_Markers      5000 non-null  object  
15  Family_History       5000 non-null  bool    
16  Country              5000 non-null  object  
17  Year                 5000 non-null  int64  
dtypes: bool(2), int64(6), object(10)
memory usage: 634.9+ KB
```

Keterangan dataframe : data memiliki 5000 bari dan 18 columns

Cleaning dataframe

1. Pengecekan dan hapus/ganti nilai Null
2. Pengecekan dan penghapusan data yang sama
3. Transformasi judul columns
4. transformasi data dengan tipe objek

1. Pengecekan dan hapus/ganto nilai null

```
In [7]: df.isnull().sum ()
```

```
Out[7]: baris                                0
Patient_ID                                0
Age                                        0
Gender                                    0
Cancer_Type                              0
Laterality                               0
Date_of_Diagnosis                         0
Stage_at_Diagnosis                        0
Treatment_Type                            0
Surgery_Status                            0
Radiation_Therapy                         0
Chemotherapy                             0
Outcome_Status                            0
Survival_Time_Months                     0
Genetic_Markers                           0
Family_History                            0
Country                                   0
Year                                       0
dtype: int64
```

2. Duplicate

```
In [8]: df.duplicated().sum()
```

```
Out[8]: np.int64(0)
```

3. judul column

```
In [9]: df.columns
```

```
Out[9]: Index(['baris', 'Patient_ID', 'Age', 'Gender', 'Cancer_Type', 'Laterality',
              'Date_of_Diagnosis', 'Stage_at_Diagnosis', 'Treatment_Type',
              'Surgery_Status', 'Radiation_Therapy', 'Chemotherapy', 'Outcome_S
tatus',
              'Survival_Time_Months', 'Genetic_Markers', 'Family_History', 'Cou
untry',
              'Year'],
              dtype='object')
```

```
In [10]: df.columns=df.columns.str.lower()
df.columns
```

```
Out[10]: Index(['baris', 'patient_id', 'age', 'gender', 'cancer_type', 'laterality',
               'date_of_diagnosis', 'stage_at_diagnosis', 'treatment_type',
               'surgery_status', 'radiation_therapy', 'chemotherapy', 'outcome_status',
               'survival_time_months', 'genetic_markers', 'family_history', 'country',
               'year'],
              dtype='object')
```

4. Ubah data object

```
In [11]: df.columns
```

```
Out[11]: Index(['baris', 'patient_id', 'age', 'gender', 'cancer_type', 'laterality',
               'date_of_diagnosis', 'stage_at_diagnosis', 'treatment_type',
               'surgery_status', 'radiation_therapy', 'chemotherapy', 'outcome_status',
               'survival_time_months', 'genetic_markers', 'family_history', 'country',
               'year'],
              dtype='object')
```

```
In [12]: columns_ = ['gender', 'cancer_type', 'laterality',
                    'date_of_diagnosis', 'stage_at_diagnosis', 'treatment_type', 'outcome_status',
                    'genetic_markers', 'country']
```

```
In [13]: for col in columns_ :
          df[col]=df[col].str.lower()
```

```
In [14]: df['gender']=np.where(
          df['gender']=='f',
          'female',
          df['gender']
          )
          df['gender']=np.where(
          df['gender']=='m',
          'male',
          df['gender']
          )
```

Delete column baris

```
In [15]: df=df.drop('baris',axis =1)
```

```
In [16]: df.head()
```

```
Out[16]:
```

	patient_id	age	gender	cancer_type	laterality	date_of_diagnosis	stage_at_diagn
0	PID00001	58	female	retinoblastoma	left	25/01/2019	sta
1	PID00002	15	other	retinoblastoma	right	21/10/2021	sta
2	PID00003	64	male	retinoblastoma	bilateral	12/03/2021	sta
3	PID00004	33	male	melanoma	right	10/05/2021	st
4	PID00005	8	other	lymphoma	left	24/11/2019	st

Analisis data

1. Analisis data central tendency
2. penyebaran data yang bersifat Numerik
3. Distribusi data dengan data character

```
In [17]: basic_analist=df.describe ()
basic_analist
```

```
Out[17]:
```

	age	radiation_therapy	chemotherapy	survival_time_months	yea
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.0000
mean	45.008000	35.447600	10.057400	60.716000	2021.5246
std	25.934387	20.565754	6.030968	34.549774	1.7162
min	1.000000	0.000000	0.000000	1.000000	2019.0000
25%	22.000000	18.000000	5.000000	31.000000	2020.0000
50%	44.000000	35.000000	10.000000	60.000000	2022.0000
75%	67.000000	53.000000	15.000000	90.000000	2023.0000
max	90.000000	70.000000	20.000000	120.000000	2024.0000

2. cek penyebaran data

```
In [18]: df.columns
```

```
Out[18]: Index(['patient_id', 'age', 'gender', 'cancer_type', 'laterality',
               'date_of_diagnosis', 'stage_at_diagnosis', 'treatment_type',
               'surgery_status', 'radiation_therapy', 'chemotherapy', 'outcome_s
               tatus',
               'survival_time_months', 'genetic_markers', 'family_history', 'cou
               ntry',
               'year'],
              dtype='object')
```

```
In [19]: columns_=['age','radiation_therapy', 'chemotherapy',
                  'survival_time_months','year']
```

```
len(columns_)
```

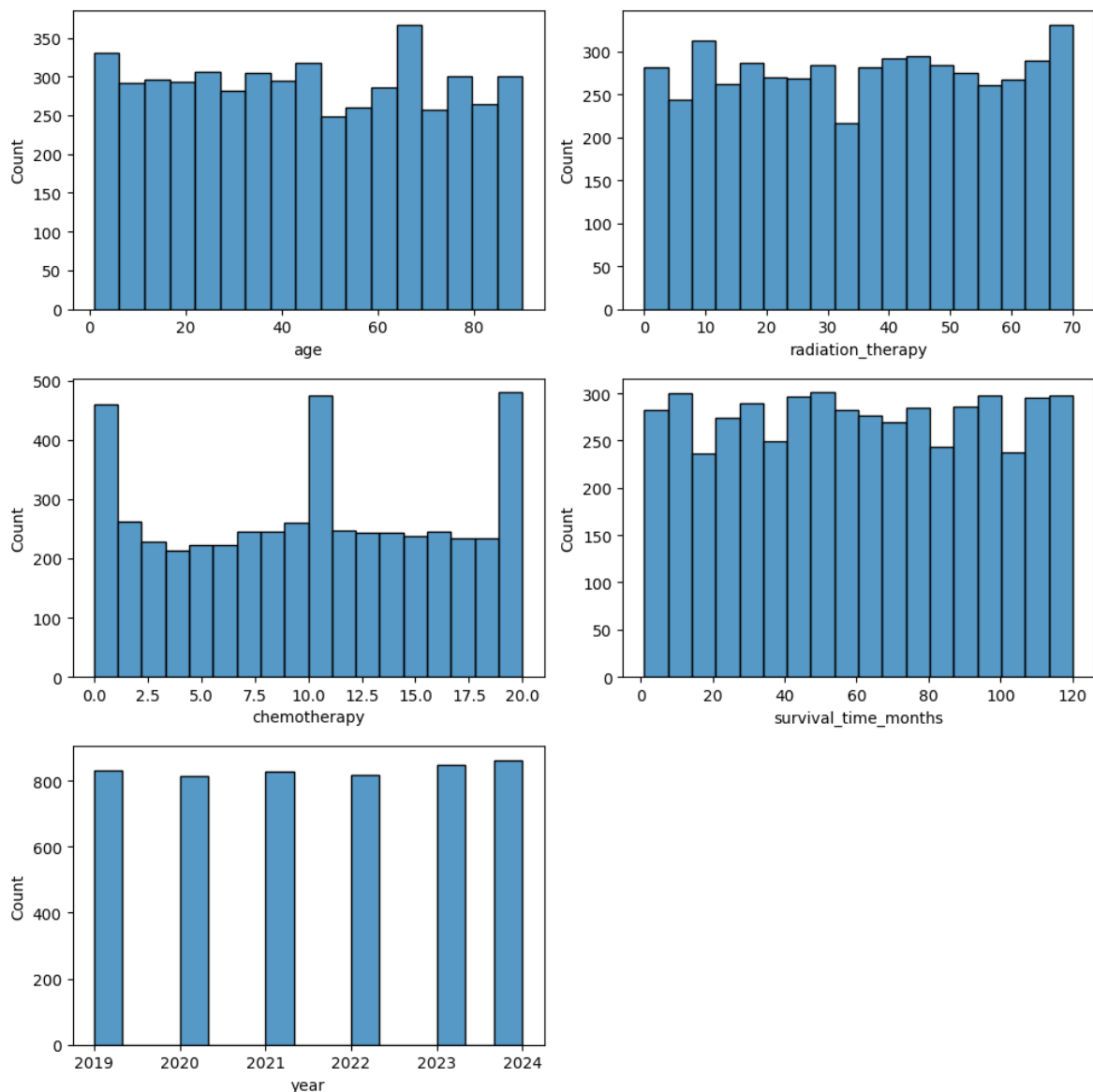
Out[19]: 5

```
In [20]: fig, ax = plt.subplots(3, 2, figsize=(10, 10))

sns.histplot(data=df, x='age', ax=ax[0, 0])
sns.histplot(data=df, x='radiation_therapy', ax=ax[0, 1])
sns.histplot(data=df, x='chemotherapy', ax=ax[1, 0])
sns.histplot(data=df, x='survival_time_months', ax=ax[1, 1])
sns.histplot(data=df, x='year', ax=ax[2, 0])

ax[2, 1].axis('off')

plt.tight_layout()
plt.show()
```



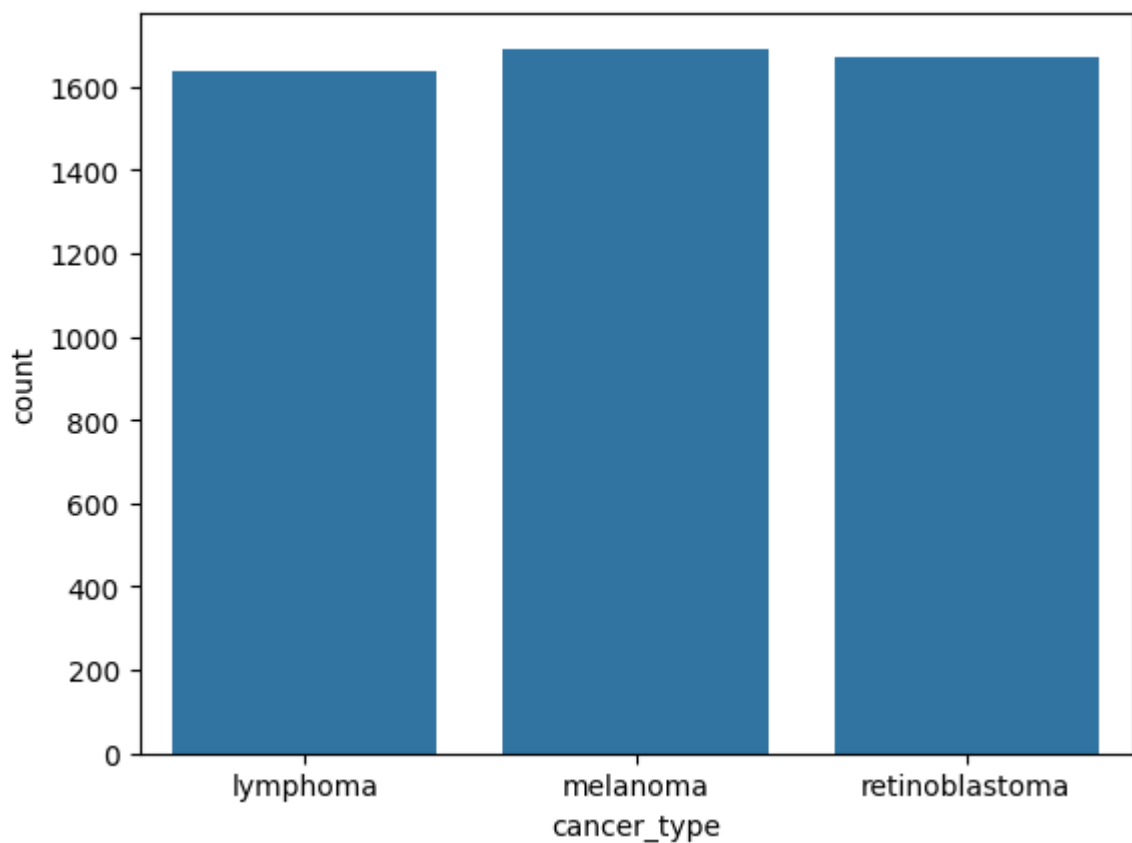
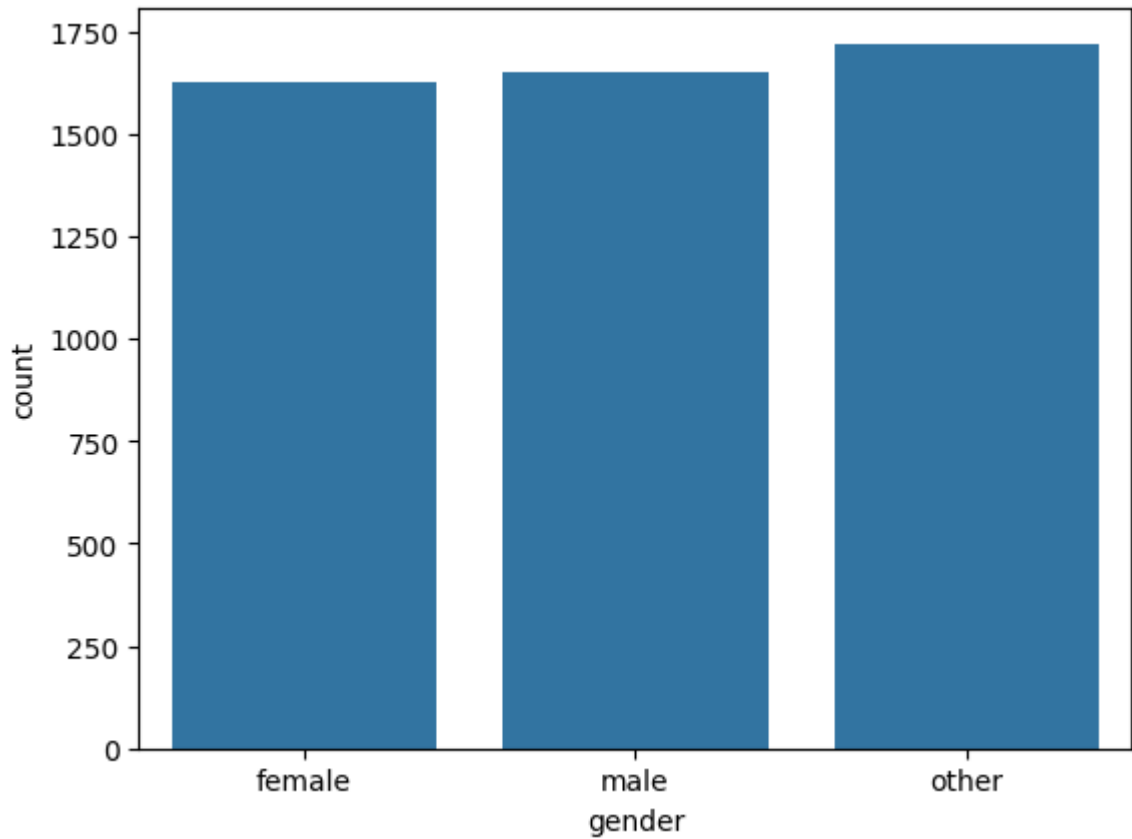
3. distribusi jumlah data pada column bertipe karakter

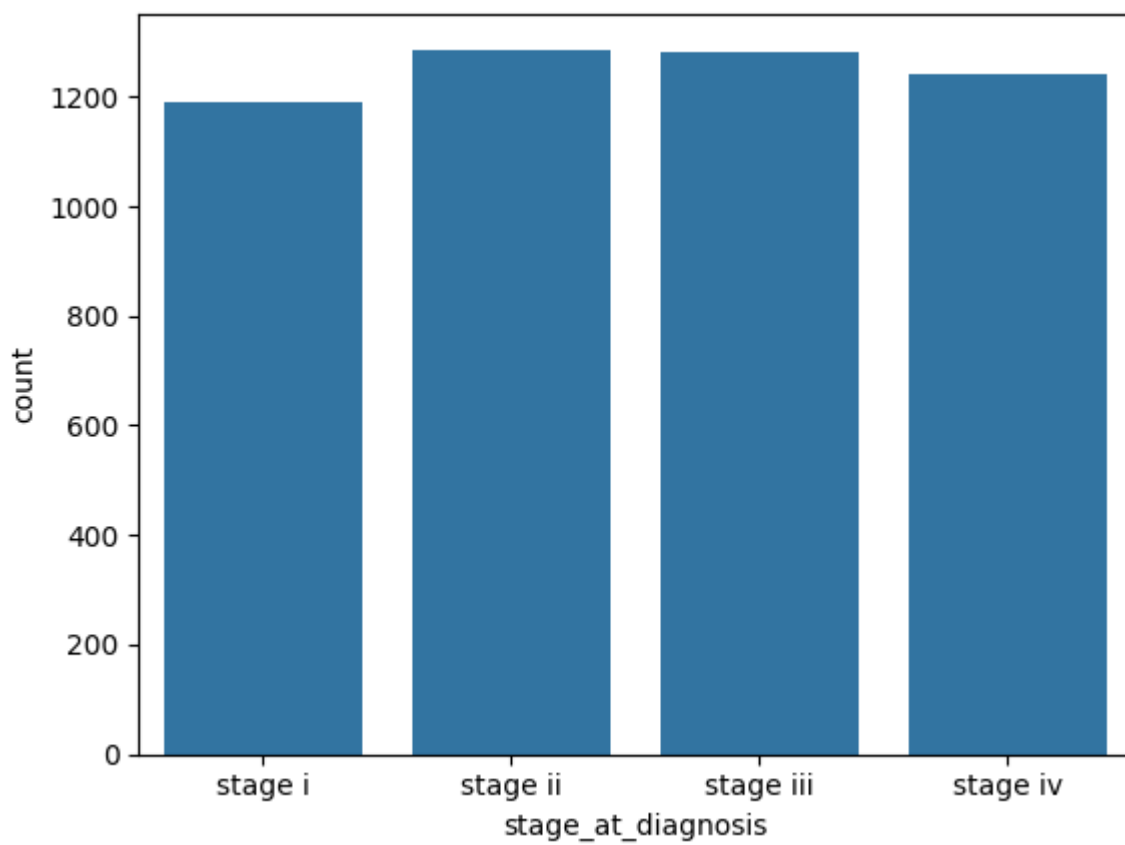
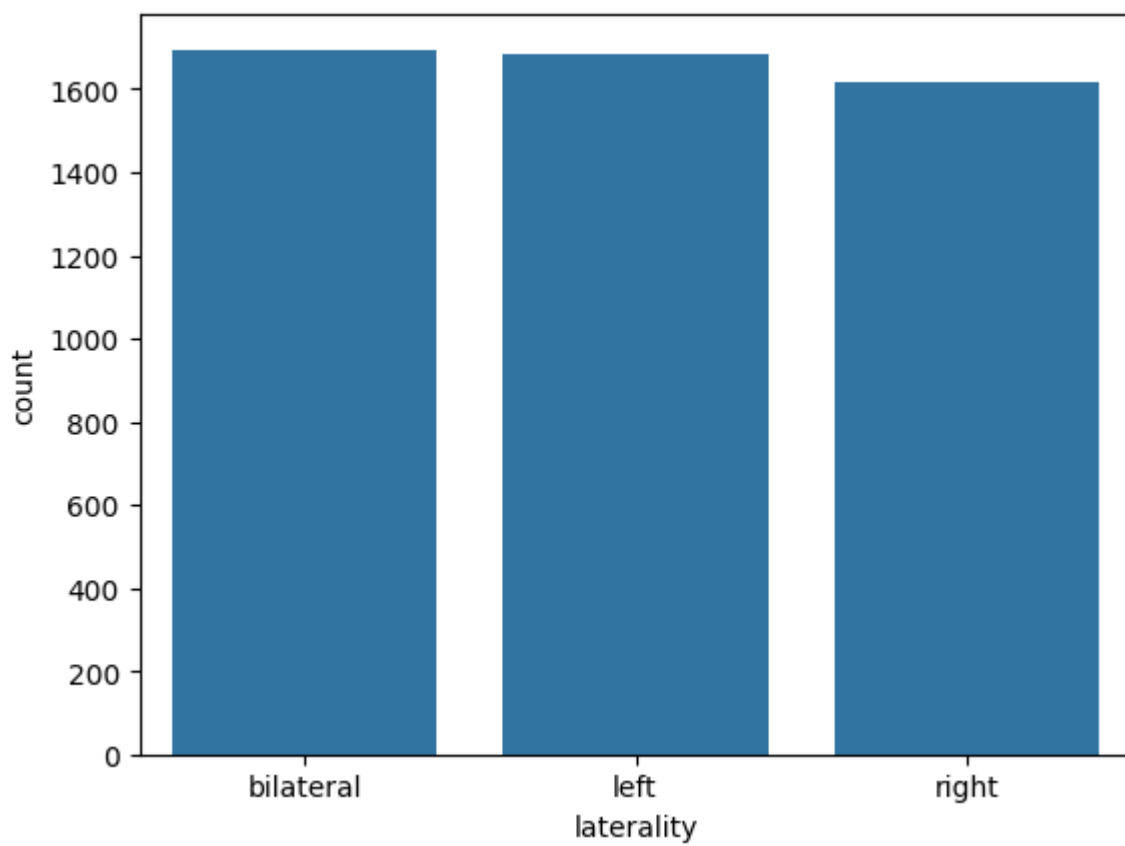
```
In [21]: columns_=[ 'gender', 'cancer_type', 'laterality',
                    'stage_at_diagnosis', 'treatment_type',
                    'outcome_status', 'genetic_markers', 'family_history', 'country'
```

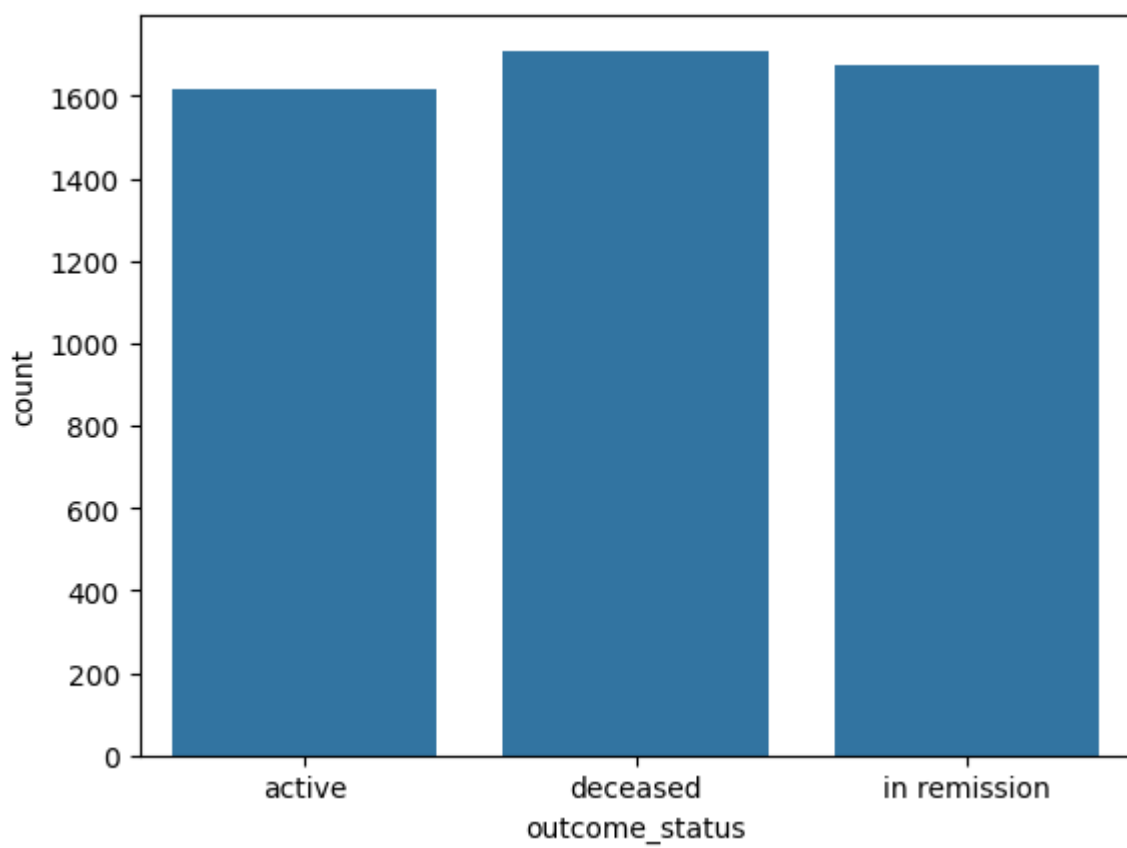
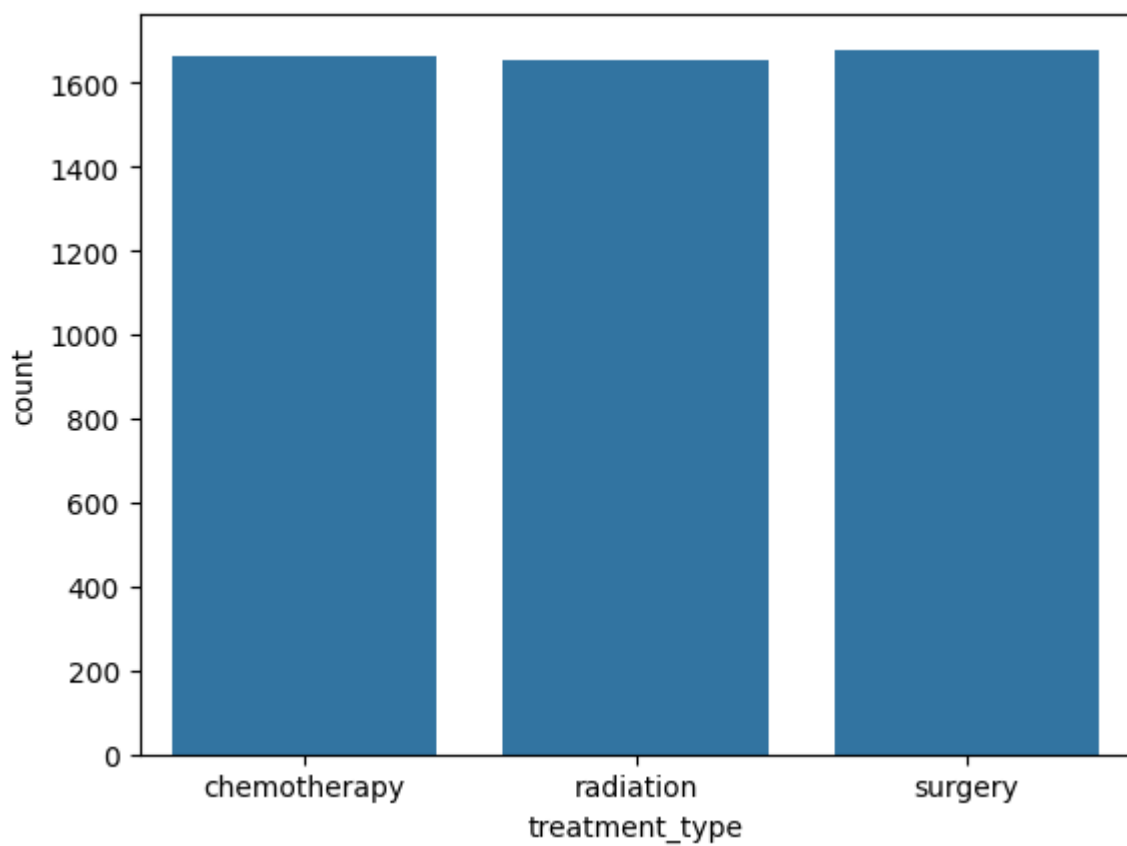
```
]
```

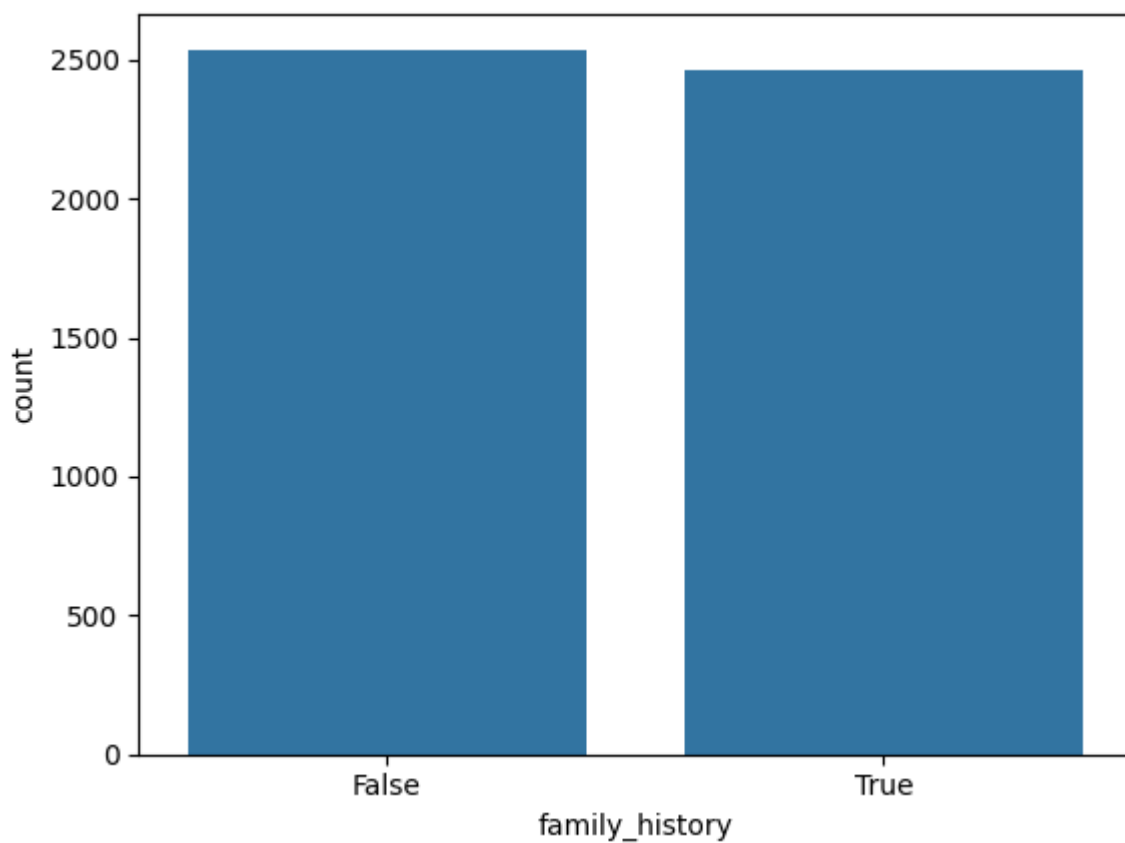
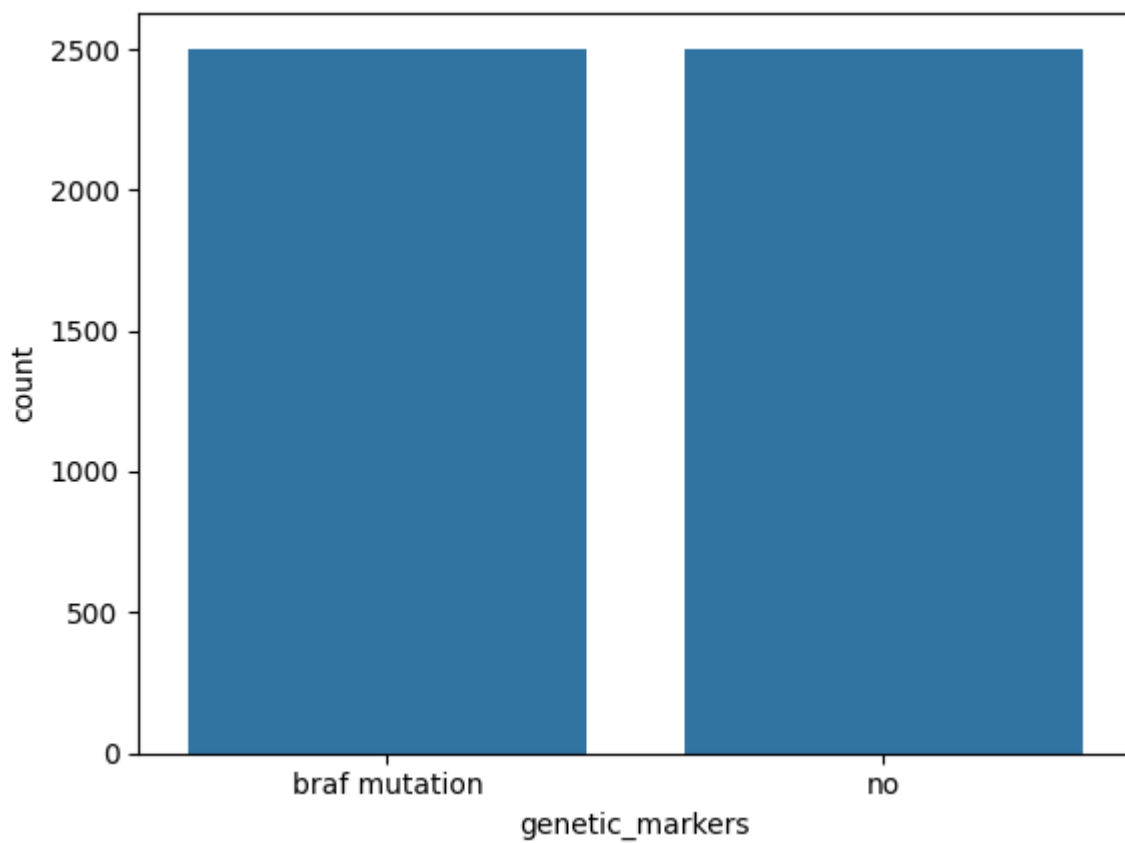
```
In [22]: for col in columns_:
          df_group = df.groupby(col).size().reset_index(name='count')
          df_group.columns = [col, 'count']

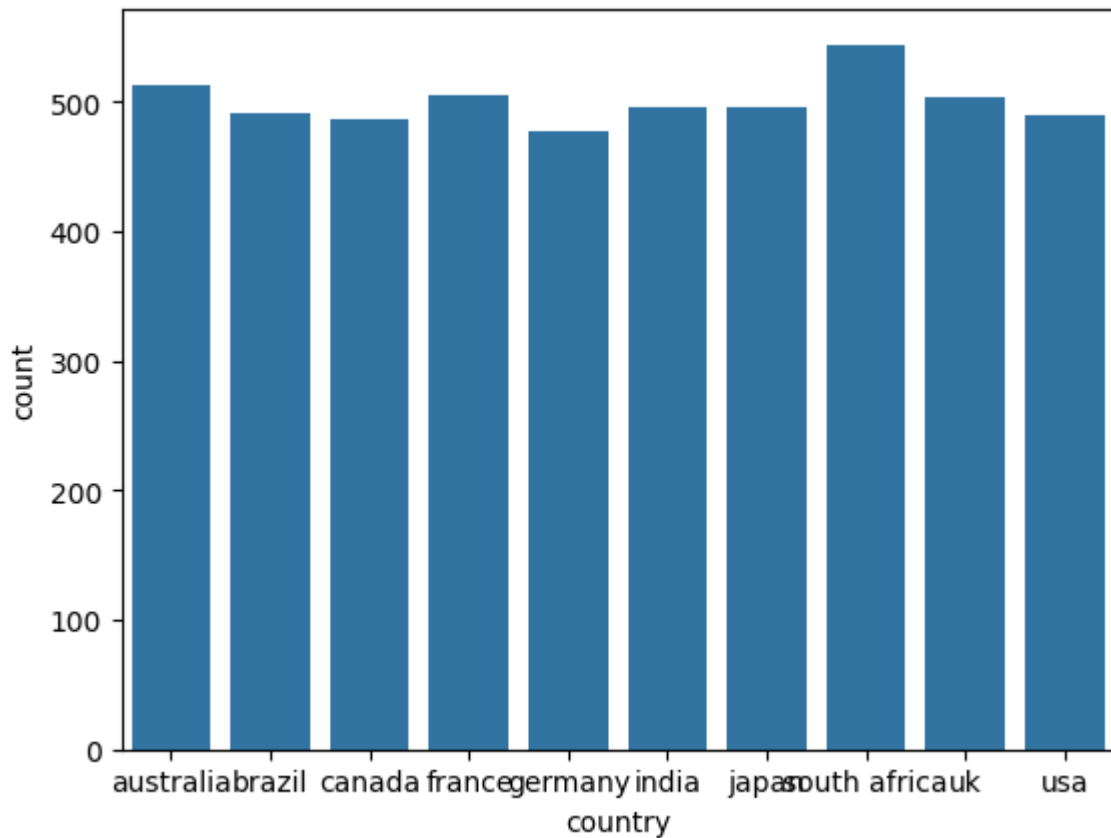
          sns.barplot(data=df_group, x=col, y='count')
          plt.show()
```











Analisis

1. Dataset terdiri dari 5.000 baris dan 18 kolom.
2. Variabel numerik menunjukkan distribusi yang relatif merata, tanpa perbedaan signifikan pada setiap rentang nilai.
3. Variabel kategorikal (seperti gender, dll.) juga memiliki proporsi yang relatif seimbang antar kategori.

Kesimpulan

1. Jumlah pasien kanker mata cenderung stabil, tidak menunjukkan tren peningkatan atau penurunan yang signifikan.
2. Faktor-faktor seperti usia, gender, dan variabel lainnya dalam dataset ini tampak memiliki kontribusi yang relatif seimbang terhadap kemungkinan terkena kanker.
3. Kemungkinan terdapat faktor lain di luar dataset yang lebih berpengaruh (misalnya: faktor genetik, lingkungan, atau gaya hidup) yang belum tercakup dalam data.